

# Exponential Periods and Factorial Moments

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## Abstract

We relate factorial moments of complex polynomials to compact exponential periods on simplices. An exact radial limit depends only on the highest two homogeneous layers and tolerates arbitrary complex lower-degree terms. Using the linear comparison theorem of Paper V, we obtain explicit families with eventually nonzero factorial moments in every dimension. One family allows every nonzero algebraic phase parameter, with degree increasing with dimension. A second family has degree four in every dimension; a characteristic-five twisted-remainder argument supplies one parameter class uniformly in the dimension. Exact coprimality calculations remove its parameter exceptions through dimension 103.

A block-radial limit gives nonflat families with arbitrarily many active angular variables and arbitrary lower-degree cross-block couplings. Algebraic specialization extends the Factorial-Conjecture conclusion to the indicated complex-parameter families; power closure gives infinitely many nonzero moments there. These conclusions are distinguished from eventual nonvanishing of moments, which follows from a nonzero limiting period. No unrestricted Factorial Conjecture or arbitrary-simplex nonvanishing theorem is asserted.

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# 1 Introduction and scope

For  $n \geq 1$  define the factorial functional

$$\mathcal{L}_n : \mathbb{C}[x_1, \dots, x_n] \longrightarrow \mathbb{C}, \quad \mathcal{L}_n(x_1^{e_1} \cdots x_n^{e_n}) = \prod_{i=1}^n e_i!.$$

Equivalently,

$$\mathcal{L}_n(f) = \int_{\mathbb{R}_{\geq 0}^n} f(x) e^{-(x_1 + \cdots + x_n)} dx. \quad (1.1)$$

The Factorial Conjecture asks whether  $\mathcal{L}_n(f^m) = 0$  for all integers  $m \geq 1$  forces  $f = 0$ ; see [2]. We say that a nonzero polynomial *satisfies the FC conclusion* when at least one of these moments is nonzero. This paper proves that conclusion for structured families, not for all polynomials in any new dimension. It does not address the consecutive-moment assertion of the Strong Factorial Conjecture.

The principal distinction is between three statements:

eventually every moment is nonzero  $\implies$  infinitely many are nonzero  $\implies$  the FC conclusion.

A nonzero radial limit proves the first. Algebraic specialization alone proves the last; power closure supplies the middle statement in our families.

**Results.** Theorem 2.1 identifies the normalized radial limit. Theorem 4.1 gives an all-dimensional family for every algebraic scale, using an interior extremum and no critical-root separation. Theorems 5.1 and 6.1 give a quartic family with a parameter class uniform in dimension. Proposition 7.2 records the exact finite exception-free range. Theorem 8.1 and Corollary 8.2 give nonflat analogues. Finally, Section 9 treats complex parameters without claiming nonzero limiting periods at every transcendental scale.

**Dependency.** Our analytic limits and finite algebraic certificates are proved here. The arithmetic nonvanishing input is the linear comparison theorem and elliptic example of Paper V [3], with the assumptions restated in Section 3. That manuscript is a working research draft; the present applications do not independently re-prove its geometric and transcendence arguments. Products of individually nonzero periods below do not require multicolumn algebraic independence or Paper IV.

Algebraic coefficients belong to  $\overline{\mathbb{Q}}$ . Write

$$S = x_1 + \cdots + x_n, \quad \Delta_{n-1} = \{t_i \geq 0 : \sum_i t_i = 1\}.$$

The measure  $dt = dt_1 \cdots dt_{n-1}$  is coordinate measure after eliminating  $t_n$ ; its volume is  $1/(n-1)!$ . A zero-dimensional simplex has mass one. This normalization is used in every slicing and limit formula.

## 2 A radial factorial-moment limit

**Theorem 2.1.** *Let  $n, D \geq 1$  and*

$$f = S^D + f_{D-1} + \cdots + f_0,$$

*where  $f_j$  is homogeneous of degree  $j$ , with arbitrary complex coefficients. Then*

$$\lim_{m \rightarrow \infty} \frac{\mathcal{L}_n(f^m)}{\Gamma(Dm + n)} = \int_{\Delta_{n-1}} \exp(f_{D-1}(t)/D) dt. \quad (2.1)$$

*If the integral is nonzero,  $\mathcal{L}_n(f^m) \neq 0$  for every sufficiently large integer  $m$ .*

*Proof.* Put  $x_i = rt_i$  in (1.1). The Jacobian is  $r^{n-1}$ . Write  $A_j(t) = f_{D-j}(t)$  and expand

$$f(rt) = r^D + A_1(t)r^{D-1} + \cdots + A_D(t).$$

For a multi-index  $\mathbf{k} = (k_1, \dots, k_D)$  set  $K = \sum_j k_j$  and  $J = \sum_j j k_j$ . After radial integration the coefficient of  $\prod_j A_j(t)^{k_j}/k_j!$  in the normalized moment is

$$C_{m,\mathbf{k}} = \frac{m!}{(m-K)!} \frac{\Gamma(Dm - J + n)}{\Gamma(Dm + n)} \quad (K \leq m),$$

and it is zero for  $K > m$ . Since  $K \leq J \leq DK \leq Dm$ , every denominator factor is a positive integer. The first  $K$  factors of the gamma ratio's denominator dominate  $m(m-1) \cdots (m-K+1)$ , and the remaining factors are at least one. Thus  $0 \leq C_{m,\mathbf{k}} \leq 1$ . For fixed  $\mathbf{k}$ ,

$$C_{m,\mathbf{k}} \sim D^{-J} m^{K-J}.$$

Only  $k_2 = \cdots = k_D = 0$  survive, and their sum is  $e^{A_1(t)/D}$ . On the compact simplex the absolute terms are bounded by the summable series  $\prod_j \|A_j\|_\infty^{k_j}/k_j!$ . Dominated summation, followed by integration, proves (2.1). A sequence converging to a nonzero complex number stays nonzero eventually.  $\square$

*Remark 2.2.* Adding  $D\beta S^{D-1}$ ,  $\beta \in \mathbb{C}$ , to the displayed family multiplies the limit by  $e^\beta$ . Multiplication of  $f$  by  $\lambda \neq 0$  multiplies its  $m$ th moment by  $\lambda^m$ . Both operations preserve the indicated nonvanishing conclusions. No sign assumption on the coefficients of  $f$  or on its lower layers is used.

## 3 The exponential-period input

For a real algebraic polynomial phase

$$q(X, Y) = \alpha X^a + \beta Y^b + q_<, \quad \gcd(a, b) = 1, \quad a, b \geq 2,$$

assume lower weighted degree for  $q_<$  and distinct nondegenerate critical values. Paper V proves irreducibility of the polynomial twisted module. For a positive polynomial domain with a strict interior phase minimum or maximum relative to the specified faces, its linear comparison theorem implies the following consequence:

$$0 \neq r \in \text{span}_{\overline{\mathbb{Q}}} \{X^i Y^j : 0 \leq i \leq a-2, 0 \leq j \leq b-2\} \implies \int_T r e^{\xi q} dX dY \notin \overline{\mathbb{Q}} \quad (\xi \in \overline{\mathbb{Q}}^\times). \quad (3.1)$$

More generally a nonzero *specialized* twisted remainder suffices. The detector may also be a saddle or the specified continuation of a density germ. Critical-difference separation and boundary-field disjointness are not needed for this linear conclusion. Its specialization uses Beukers' Theorem 3.2 [1]; these stronger hypotheses are used only for algebraic independence over the boundary field in Paper V.

For the elliptic phase used below, Paper V's continued-density example provides the same conclusion on the fixed triangle even though its real saddle is outside that triangle. Only the specified faces are used. The stronger elliptic independence theorem also implies transcendence over their value field when the interior remainder is nonzero, but absolute nonvanishing is sufficient for all moment arguments here.

## 4 Every algebraic scale in every dimension

Fix  $n \geq 3$  and choose odd  $a \geq \max(3, n-1)$ . Put

$$q_a(X, Y) = X^a - aX + Y^2, \quad T = \{0 \leq X \leq 2, |Y| \leq 1 - X/2\}.$$

Its critical points satisfy  $X^{a-1} = 1$ ,  $Y = 0$ , and their values  $-(a-1)X$  are distinct. On  $T$  the minimum is uniquely attained at  $(1, 0)$ , strictly below every boundary value. Thus (3.1) applies to  $X^{n-3}$ , whose exponent is at most  $a-2$ .

**Theorem 4.1.** *Let  $\sigma = x_3 + \cdots + x_n$ ,  $D \geq a+1$ ,  $\xi \in \overline{\mathbb{Q}}^\times$ , and  $\deg R \leq D-2$ , with arbitrary complex coefficients in  $R$ . Define*

$$f = S^D + D\xi(2^a \sigma^a S^{D-1-a} - 2a\sigma S^{D-2} + (x_2 - x_1)^2 S^{D-3}) + R. \quad (4.1)$$

Then

$$\lim_{m \rightarrow \infty} \frac{\mathcal{L}_n(f^m)}{\Gamma(Dm+n)} = \frac{1}{2^{n-1}(n-3)!} \int_T X^{n-3} e^{\xi q_a(X,Y)} dX dY \neq 0.$$

Consequently every such  $f$  has eventually nonzero factorial moments. At  $\xi = 0$  the same conclusion holds with limit  $1/(n-1)!$ .

*Proof.* Write  $u = t_1$ ,  $v = t_2$ . Integrating the remaining  $n-2$  simplex coordinates gives the weight  $(1-u-v)^{n-3}/(n-3)!$  on  $u, v \geq 0$ ,  $u+v \leq 1$ . The affine map

$$X = 2(1-u-v), \quad Y = v-u$$

has absolute Jacobian four and maps this triangle onto  $T$ . Thus

$$\begin{aligned} & \int_{\Delta_{n-1}} e^{\xi q_a(2 \sum_{j \geq 3} t_j, t_2 - t_1)} dt \\ &= \frac{1}{2^{n-1}(n-3)!} \int_T X^{n-3} e^{\xi q_a(X,Y)} dX dY. \end{aligned}$$

The homogeneous layer of degree  $D-1$  in (4.1), divided by  $D$  on  $S=1$ , is exactly this phase. Apply Theorem 2.1 and (3.1). At zero the integrand is one.  $\square$

For  $a \geq 5$  the critical values contain two different opposite pairs, so ordered critical differences are not separated. This family therefore uses the weaker linear theorem, not the full-independence theorem disguised as a scalar corollary. The price of allowing every algebraic scale uniformly is increasing degree as  $n$  increases.

## 5 A characteristic-five theorem for the elliptic family

Take

$$q(X, Y) = Y^2 - X^3 + 3X, \quad T = \text{conv}\{(-\frac{1}{2}, -1), (\frac{1}{2}, 0), (-\frac{1}{2}, 1)\}, \quad \ell = \frac{1 - 2X - 2Y}{4}.$$

The form  $\ell$  is a barycentric coordinate on this triangle. Put

$$J_k(z) = \int_T \ell^k e^{zq} dX dY, \quad M_0 = \int_T e^{zq}, \quad M_1 = \int_T X e^{zq}.$$

For  $h = 1/z$  the twisted operators are

$$D_X = h\partial_X + 3(1 - X^2), \quad D_Y = h\partial_Y + 2Y.$$

Their additive quotient has basis  $1, X$  over  $\mathbb{Z}[1/6, h]$ . Write  $N_h(r) = a_r(h) + b_r(h)X$  for the unique remainder. The reductions are

$$[X^u Y^v] = [X^{u-2} Y^v] + \frac{h(u-2)}{3} [X^{u-3} Y^v] \quad (u \geq 2), \quad (5.1)$$

$$[X^u Y^v] = -\frac{h(v-1)}{2} [X^u Y^{v-2}] \quad (v \geq 1), \quad (5.2)$$

where zero-coefficient terms are omitted. In particular odd  $Y$  powers reduce to zero. The two reductions commute, and their leading coefficients are units; the resulting normal form remains free after reduction modulo five. This is an additive quotient by twisted derivatives, *not* a quotient algebra.

Set  $a_k = a_{\ell^k}$ ,  $b_k = b_{\ell^k}$ . Literal Stokes gives

$$J_k(\xi) = a_k(1/\xi)M_0(\xi) + b_k(1/\xi)M_1(\xi) + B_k(\xi), \quad (5.3)$$

where  $B_k(\xi)$  is in the fixed boundary span. A nonzero remainder vector therefore implies  $J_k(\xi) \neq 0$  by Paper V.

**Theorem 5.1** (Uniform nonexactness in characteristic five). *Let  $\xi \in \overline{\mathbb{Q}}^\times$ . Suppose that at some nonarchimedean place above five,*

$$v(\xi) \leq 0. \quad (5.4)$$

*Then  $(a_k(1/\xi), b_k(1/\xi)) \neq (0, 0)$  and  $J_k(\xi) \neq 0$  for every integer  $k \geq 0$ .*

*Proof.* In characteristic five, multiplication by  $\ell^5$  commutes with both twisted operators because its two partial derivatives vanish. Unlike multiplication by  $\ell$ , it descends to the additive quotient. In the basis  $1, X$ , reductions (5.1)–(5.2) give

$$A(h) = \begin{pmatrix} h-1 & 2h^2+2 \\ 2 & -h-1 \end{pmatrix}, \quad \det A(h) = 2 \quad \text{in } \mathbb{F}_5[h].$$

For  $v_k = (a_k, b_k)^T$  modulo five the first five vectors are

$$\begin{aligned} v_0 &= (1, 0)^T, & v_1 &= (-1, 2)^T, & v_2 &= (-2h, 1)^T, \\ v_3 &= (2h+2, -2h-1)^T, & v_4 &= (2h^2+h+1, 2h)^T. \end{aligned}$$

Each is nonzero at every specialization over every extension of  $\mathbb{F}_5$ . For  $v_3$  its coordinates sum to one; for  $v_4$ , the first plus  $(2 - h)$  times the second is one modulo five. For  $k = 5b + r$ ,  $0 \leq r < 5$ ,

$$v_k(h) = A(h)^b v_r(h).$$

Invertibility proves the assertion for every exponent in characteristic five. No extrapolation from a finite scan is involved.

Choose a number field containing  $\xi$  and the stated valuation. Its element  $h = 1/\xi$  is integral at that place, as are the remainder coefficients. If both specialized coefficients were zero, reduction in the residue field would contradict the preceding identity. Finally apply (5.3) and the scalar comparison input.  $\square$

Examples include  $\xi = i$ ,  $101i$ , and  $Ni$  for any integer  $N$  not divisible by five. Thus the result includes arbitrarily large purely imaginary phases; it is not limited to a small-oscillation positivity range.

## 6 Degree four in every dimension

Define the cubic homogeneous polynomial

$$H_n = S(x_1 + 2x_2 - S)^2 - (x_1 - S/2)^3 + 3S^2(x_1 - S/2). \quad (6.1)$$

**Theorem 6.1.** *Let  $n \geq 3$ ,  $D \geq 4$ , and let  $\xi \in \overline{\mathbb{Q}}^\times$  satisfy (5.4). For any complex polynomial  $R$  of degree at most  $D - 2$ , put*

$$f = S^D + D\xi S^{D-4}H_n + R. \quad (6.2)$$

Then

$$\lim_{m \rightarrow \infty} \frac{\mathcal{L}_n(f^m)}{\Gamma(Dm + n)} = \frac{J_{n-3}(\xi)}{2(n-3)!} \neq 0. \quad (6.3)$$

In particular  $D = 4$  and the same  $\xi = 101i$  work in every dimension.

*Proof.* Write  $Q(u, v) = q(u - 1/2, u + 2v - 1)$ , so  $H_n = S^3 Q(x_1/S, x_2/S)$ . Simplex slicing gives

$$\int_{\Delta_{n-1}} e^{\xi Q(t_1, t_2)} dt = \frac{1}{(n-3)!} \int_{u, v \geq 0, u+v \leq 1} (1 - u - v)^{n-3} e^{\xi Q(u, v)} du dv.$$

The map  $(X, Y) = (u - 1/2, u + 2v - 1)$  has positive Jacobian two, maps the standard triangle onto  $T$ , and takes  $1 - u - v$  to  $\ell$ . This gives the right side of (6.3). Theorem 2.1 supplies the limit and Theorem 5.1 makes it nonzero.  $\square$

For example

$$f = S^4 + 404i(S(x_1 + 2x_2 - S)^2 - (x_1 - S/2)^3 + 3S^2(x_1 - S/2)) + R_{\leq 2}$$

has eventually nonzero factorial moments for every  $n \geq 3$ . The quadratic perturbation may involve every variable with arbitrary complex coefficients. These are not merely polynomials in three variables with unused coordinates adjoined.

## 7 Finite exceptions and an exact recurrence

For a fixed exponent define

$$E_k = \{\xi \in \overline{\mathbb{Q}}^\times : a_k(1/\xi) = b_k(1/\xi) = 0\}.$$

This is finite. At  $h = 0$  the quotient is reduction modulo  $X^2 - 1, Y$ , and  $\ell(1, 0) = -1/4$ ,  $\ell(-1, 0) = 3/4$ ; hence its remainder vector is not identically zero for any exponent. Every parameter outside  $E_k$  has a nonzero specialized remainder, whether or not it satisfies the valuation test. Any member of  $E_k$  must be strictly divisible by five at every place above five. Membership in  $E_k$  only defeats this interior-remainder certificate: it does not prove  $J_k(\xi) = 0$ .

**Proposition 7.1** (Integer recurrence). *Set*

$$W_k(t) = 4^k(a_k(3t), b_k(3t)), \quad W_0 = (1, 0), \quad W_1 = (1, -2).$$

*Its coordinates are in  $\mathbb{Z}[t]$ , and*

$$\begin{aligned} W_{k+2} = & 2W_{k+1} + [3 - 6(2k+1)t]W_k \\ & + 4ktW_{k-1} - 36k(k-1)t^2W_{k-2}, \end{aligned} \tag{7.1}$$

*with every zero-prefactor term omitted.*

*Proof.* Work coefficientwise in the formal series  $F(s) = N_h(e^{s\ell})$ . Twisted integration by parts gives

$$N_h(Ye^{s\ell}) = \frac{hs}{4}F, \quad N_h(X^2e^{s\ell}) = (1 - hs/6)F, \quad N_h(Y^2e^{s\ell}) = (h^2s^2/16 - h/2)F.$$

Also  $N_h(XYe^{s\ell}) = (hs/4)N_h(Xe^{s\ell})$  and  $N_h(Xe^{s\ell}) = (1/2 - hs/4)F - 2F'$ . Substituting into  $F''$  yields

$$F'' = \left(\frac{1}{2} - \frac{hs}{4}\right)F' + \left(\frac{3}{16} - \frac{h}{8} + \frac{hs}{48} - \frac{h^2s^2}{64}\right)F.$$

Equate exponential-generating coefficients, put  $h = 3t$ , and multiply by  $4^{k+2}$ . This is (7.1); integrality follows by induction.  $\square$

**Proposition 7.2** (Finite exact verification). *For  $0 \leq k \leq 100$  the two coordinates of  $W_k$  have gcd one over  $\mathbb{Q}[t]$ . Thus  $E_k = \emptyset$  in this range. Consequently Theorem 6.1 holds for every nonzero algebraic  $\xi$  when  $3 \leq n \leq 103$ . At  $\xi = 0$  its limit is the simplex volume.*

*Proof.* The supplied script constructs (7.1) in exact integer polynomial arithmetic and performs the 101 Euclidean gcd computations. All gcd degrees are zero; the maximum coordinate degree is 50. It also compares the recurrence against independent monomial Stokes reduction through  $k = 12$ . The recurrence was proved above, so the gcd calculation is a finite reproducible certificate for exactly the stated range. A constant gcd excludes simultaneous roots over  $\mathbb{C}$ , giving the consequence.  $\square$

No uniform coprimality assertion for every  $k$  is inferred from this test. The all-exponent theorem is instead the characteristic-five argument of Theorem 5.1.

## 8 Block-radial limits and nonflat families

Partition the variables into nonempty blocks of sizes  $n_j$ ,  $1 \leq j \leq r$ , and let  $S_j$  denote their sums. Fix integers  $d_j \geq 1$ , and put

$$F_0 = \prod_{j=1}^r S_j^{d_j}, \quad D = \sum_j d_j, \quad C_d = \prod_j d_j^{d_j}.$$

A block angular variable  $t^{(j)}$  belongs to  $\Delta_{n_j-1}$ .

**Theorem 8.1** (Block-radial limit). *For a homogeneous  $G$  of degree  $D - 1$  and  $\deg R \leq D - 2$ , both with arbitrary complex coefficients, let  $f = F_0 + G + R$ . Then*

$$\lim_{m \rightarrow \infty} \frac{\mathcal{L}_n(f^m)}{\prod_j \Gamma(d_j m + n_j)} = \int_{\prod_j \Delta_{n_j-1}} \exp\left(\frac{G(d_1 t^{(1)}, \dots, d_r t^{(r)})}{C_d}\right) \prod_j dt^{(j)}. \quad (8.1)$$

A nonzero integral again gives eventual nonvanishing.

*Proof.* Use block polar coordinates  $x^{(j)} = s_j t^{(j)}$ , whose Jacobian is  $\prod_j s_j^{n_j-1}$ , and set  $s_j = mu_j$ . On a fixed compact neighborhood  $V$  of  $d = (d_1, \dots, d_r)$  inside the positive orthant,

$$m^{-D} f(mu_1 t^{(1)}, \dots, mu_r t^{(r)}) = \prod_j u_j^{d_j} \left( 1 + \frac{G(u_1 t^{(1)}, \dots, u_r t^{(r)})}{m \prod_j u_j^{d_j}} + O(m^{-2}) \right),$$

uniformly in  $u \in V$  and the angular variables. Raising the relative factor to the  $m$ th power converges uniformly to the corresponding exponential. Independent gamma random variables of shapes  $d_j m + n_j$ , after division by  $m$ , converge in probability to  $d_j$  (their means are  $d_j + n_j/m$  and variances are  $d_j/m + n_j/m^2$ ). Thus the part on  $V$  has the required limit.

We must control the complement without dividing by  $F_0$  near a coordinate boundary. Put  $U = \sum_j u_j$  and

$$H_m(u, t) = e^{-U} |m^{-D} f(mu_1 t^{(1)}, \dots, mu_r t^{(r)})|.$$

On every compact radial set, including its coordinate boundary,  $H_m$  converges uniformly to  $H_0(u) = e^{-U} \prod_j u_j^{d_j}$ . This function has a unique global maximum  $M = e^{-D} C_d$  at  $d$ , and vanishes at the coordinate boundary. Hence on a compact complement of  $V$  its maximum is at most  $M e^{-\eta}$  for some  $\eta > 0$ , for all sufficiently large  $m$ .

Uniformly for  $m \geq 1$  there is a constant  $C$  with  $H_m \leq C e^{-U} (1+U)^D$ . For sufficiently large  $U$  this is bounded by  $M e^{-\eta} e^{-U/2}$ . The resulting tail integral of  $H_m^m \prod_j u_j^{n_j-1}$  is at most a constant times  $M^m e^{-\eta m} m^{-n}$ , where  $n = \sum_j n_j$ . By Stirling's formula the normalization  $\prod_j \Gamma(d_j m + n_j) / m^{Dm+n}$  is  $M^m$  times a nonzero constant and  $m^{-r/2}(1+o(1))$ . Both complementary regions are therefore exponentially negligible after normalization. This proves the limit and justifies the local gamma argument without an unsafe denominator estimate on the boundary.  $\square$

**Corollary 8.2** (Several active angular blocks). *For each block choose a polynomial  $Q_j$  of degree  $a_j \leq d_j - 1$  and a scale  $\xi_j$  such that*

$$I_j = \int_{\Delta_{n_j-1}} e^{\xi_j Q_j(t)} dt \neq 0.$$

Let  $\tilde{Q}_j(x^{(j)}) = S_j^{a_j} Q_j(x^{(j)}/S_j)$ , and set

$$G = \sum_j d_j \xi_j \left( \prod_{l \neq j} S_l^{d_l} \right) S_j^{d_j - 1 - a_j} \tilde{Q}_j(x^{(j)}).$$

For every  $\deg R \leq D - 2$ , the polynomial  $f = F_0 + G + R$  has limiting ratio  $\prod_j I_j \neq 0$  in (8.1). In particular the odd-degree and elliptic constructions may be assigned independently to arbitrarily many blocks. Inactive singleton blocks contribute mass one.

*Proof.* At  $x^{(j)} = d_j t^{(j)}$ , the quotient  $G/C_d$  is  $\sum_j \xi_j Q_j(t^{(j)})$ . The integral in (8.1) factors.  $\square$

The top  $F_0$  is genuinely nonflat when there are at least two blocks. Lower-degree cross-block couplings in  $R$  are unrestricted. The degree- $D - 1$  correction in this corollary is deliberately block-separable; arbitrary coupled angular phases at that degree would require a new nonvanishing argument. No several-phase comparison theorem is used to multiply the individual nonzero numbers.

## 9 Complex parameters and power closure

**Lemma 9.1** (Algebraic specialization of moment equations). *Let a polynomial family  $f_\theta$  be defined over  $\mathbb{Q}$  on an affine finite-type parameter space, and let  $U$  be a Zariski-open subset defined over  $\mathbb{Q}$ . If no  $\mathbb{Q}$ -point of  $U$  has all its positive factorial moments zero, then no  $\mathbb{C}$ -point of  $U$  does.*

*Proof.* A hypothetical complex point lies in a principal affine open contained in  $U$ . Adjoin an inverse to its defining nonvanishing polynomial. The moment polynomials generate an ideal in this finitely generated  $\mathbb{Q}$ -algebra. The complex point makes it proper. The weak Nullstellensatz gives a  $\mathbb{Q}$ -valued point of its zero set, at which every moment still vanishes and the same open condition holds, a contradiction. No finite uniform moment bound is required: Noetherianity applies to the ideal of all moments.  $\square$

**Theorem 9.2.** *The odd-degree family (4.1) satisfies the FC conclusion for every  $\xi \in \mathbb{C}$ . The elliptic family (6.2) satisfies it for every  $\xi \in \mathbb{C} \setminus E_{n-3}$ , with zero included. For  $3 \leq n \leq 103$  it therefore holds for every complex scale. In all these families there are infinitely many nonzero factorial moments. Analogous conclusions hold for the block-separable families when each scale lies outside its indicated finite algebraic exception set.*

*Proof.* For the odd-degree family all algebraic scales give a nonzero limit by Theorem 4.1, including zero. Apply Lemma 9.1, allowing the coefficients of  $R$  as additional parameters. For the elliptic family at a nonzero scale outside  $E_{n-3}$ , at least one specialized remainder coordinate is nonzero. After clearing powers of  $\xi$ , invert that coordinate and  $\xi$ . At every algebraic point in this open set, scalar comparison and Theorem 2.1 give a nonzero limit. The lemma applies. At  $\xi = 0$  the radial limit is directly the simplex volume, for arbitrary complex  $R$ . The block argument uses the product of the selected nonvanishing conditions and Corollary 8.2.

These families are closed under positive powers with the same angular phase and scales. For a flat top,

$$(S^D + f_{D-1} + R)^N = S^{DN} + N S^{D(N-1)} f_{D-1} + R', \quad \deg R' \leq DN - 2.$$

Thus  $D$  changes to  $DN$  while  $f_{D-1}/D$  on  $S = 1$  is unchanged. For the block family  $d_j$  changes to  $N d_j$  and  $G$  to  $N F_0^{N-1} G$ , preserving the same block angular correction. If only finitely many

moments of  $f$  were nonzero, choose  $N$  larger than all their indices. Then every positive moment of  $f^N$  would vanish, contradicting its FC conclusion.  $\square$

*Remark 9.3.* The specialization theorem does not show that the limiting exponential period is nonzero for every transcendental scale. Its conclusion concerns the polynomial moment equations. Eventual nonvanishing is asserted where a nonzero limit was proved; infinitely many nonzero moments is the conclusion obtained from specialization and power closure alone.

## 10 General transfers, verification, and remaining problems

**Proposition 10.1** (A finite exceptional set for each slicing exponent). *Suppose a triangular phase satisfies Paper V’s scalar comparison and has a Laurent Stokes basis with  $h = 1/z$  reduction regular at  $h = 0$ , specializing to its Jacobian quotient. Let  $\ell$  be a barycentric linear form which is nonzero at at least one critical point. For each  $k \geq 0$ , the integral  $\int_T \ell^k e^{\xi q}$  is nonzero for all but finitely many  $\xi \in \overline{\mathbb{Q}}^\times$ .*

*Proof.* The Jacobian class of  $\ell^k$  is nonzero, by evaluation at that critical point. Thus its polynomial remainder vector is not identically zero in  $h$ . Outside the finite common-zero set its specialization is nonzero; apply scalar comparison. A hypothetical exceptional parameter would only be a failure of this certificate, not a proven zero of the integral.  $\square$

The characteristic-five theorem adds a uniform arithmetic certificate for all exponents to this finite-exception argument. More generally, in good characteristic  $p$ , multiplication by a  $p$ th power commutes with the twisted derivatives. An invertible induced matrix and nonzero initial vectors for exponents  $0, \dots, p-1$  give an all-exponent nonexactness test. It must be proved in the actual free twisted quotient; multiplication by an arbitrary amplitude need not descend.

**Verification.** Run the exact script from the repository root:

```
python3 scripts/check_comparison_factorial.py \
    --max-k 100 --output .build/comparison-factorial.json
```

It checks the two-coordinate mod-five matrix, its unit determinant and five initial vectors, the integer recurrence against direct normal forms, all 101 finite gcds, affine Jacobians and low-degree normalization identities, and related Paper V examples. The tests use exact arithmetic, not oscillatory quadrature. The finite range in Proposition 7.2 is computational; Theorem 5.1 is an all-exponent proof.

**Boundaries.** The explicit families do not establish unrestricted  $\text{FC}(n)$ , arbitrary flat-top phases, or unrestricted simplex exponential nonvanishing. Neither does a zero leading radial limit disprove FC: higher asymptotic coefficients may be nonzero. Uniform emptiness of every  $E_k$  and nonvanishing for coupled angular phases remain separate targets. The companion simplex candidate [4] records these distinctions and a counterexample to the overly broad positive-amplitude induction. The multicolumn project is separate from the product argument of Corollary 8.2.

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- [4] C. D. Long and A.-A. Le Blanc, *Simplex Exponential Periods: Reductions and Open Comparison Problems*, unnumbered candidate note, October 2026, in the same repository.